Dániel Bacsó (N6M007) - Homework2

Neffke, Otto and Weyh (2017) address the structure of labour mobility across industries and determining whether human capital has a strong industry-specific component. Earlier relatedness measures (co-location, co-exports, input-output links) infer skill similarity only indirectly, capturing geography, shared inputs, or value-chain links, rather than whether workers can actually transfer their skills between industries. The study therefore asks whether inter-industry labour flows can reveal an underlying, industry-specific component of human capital. Using German social security data (1999-2008), it shows that flows are highly structured, stable over time, similar for local and long-distance moves and across occupations, and a stronger predictor of regional industry growth and entry than co-location or input-output measures.

Sánchez-Moral et al. (2022) apply the labour-flow-based skill-relatedness approach building on Neffke et al. (2017) to Spain and, using the resulting industry relatedness matrix, construct a provincial-level related-variety index. They then examine how this index relates to regional performance during the crisis (2009-2013) and the recovery (2014-2018) periods, finding that related variety is linked to lower unemployment growth in the crisis and to GDP growth in the recovery.

Wang et al. (2026) extend Neffke et al. (2017) by measuring skill relatedness from UK job-posting skill demand (technical-digital and social-cognitive) rather than labour flows, and relating it to manufacturing entry and exit across four types of UK regions. Higher relatedness is associated with more entry and less exit, especially when both skill types are combined, and the technical-digital effect on entry is about twice as strong in left-behind regions.

Research Gap: Previous research shows that labour mobility between industries depends heavily on skill-relatedness, but we do not know how well skills from declining, fossil-fuel-intensive industries transfer to emerging green technology sectors. This is important because a significant skills mismatch that arises during the green transition could lead to structural unemployment and delay the realization of regional decarbonisation targets.

Research question: How skill-related are traditional carbon-intensive manufacturing industries to emerging green energy sectors, and which regional skill overlaps best facilitate a smooth green transition?

Data and potential users: A possible source could be skill requirements in job advertisements for green and brown sectors. Regional economic development decision-makers would use the findings on retraining programmes.